

ENGINE LEVEL

is the geometry right?

- **Line-of-sight engine**
profile sampling · curvature + refraction
5 switchable defects, injected one at a time
- **Benchmark**
15 terrain cases, expectations derived
from the critical distance, not intuition
4 behavioural properties, metamorphic
- **Mutation analysis**
the whole benchmark, run against
each deliberately broken engine
- **4 defects killed · 1 equivalent**
equivalence proven by a paired
sweep, not assumed

DESIGN LEVEL

is the inference right?

- **Synthetic landscape**
anisotropic ridges · a lake flooded to a
constant, exactly as real DEMs do
- **Ground truth by construction**
S0 exchangeable → p exactly uniform
S1 best of K → known theoretical
power, ceiling 85 %
- **Paired contrasts**
correct · water unmasked · tight crop
same data — only the defect differs
- **Calibration · power · mechanism**
each defect measured against an
exact expectation, not an assumption

DIAGNOSTICS · ANY REAL STUDY

D1 null placements touching water = 0

D2 orientation coverage = 100 %

Only a validated engine and a diagnosed design touch real terrain.